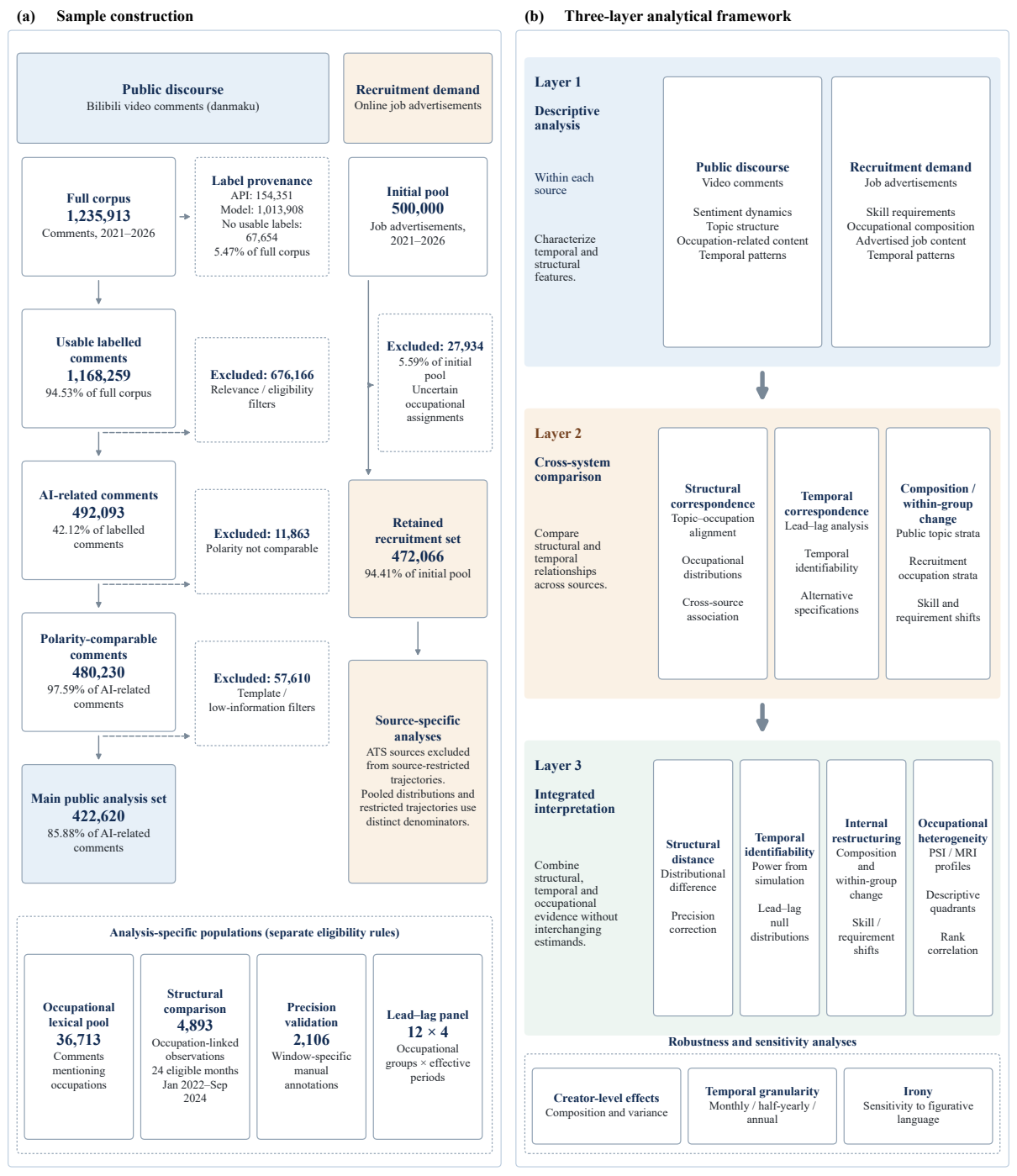

(a) Sample construction
Public discourse
Bilibili video comments (danmaku)
Recruitment demand
Online job advertisements
Full corpus
1,235,913
Comments, 2021–2026
Label provenance
API: 154,351
Model: 1,013,908
No usable labels: 67,654
5.47% of full corpus
Initial pool
500,000
Job advertisements, 2021–2026
Usable labelled comments
1,168,259
94.53% of full corpus
Excluded: 676,166
Relevance / eligibility filters
Excluded: 27,934
5.59% of initial pool
Uncertain occupational assignments
AI-related comments
492,093
42.12% of labelled comments
Excluded: 11,863
Polarity not comparable
Retained recruitment set
472,066
94.41% of initial pool
Polarity-comparable comments
480,230
97.59% of AI-related comments
Excluded: 57,610
Template / low-information filters
Source-specific analyses
ATS sources excluded from source-restricted trajectories.
Pooled distributions and restricted trajectories use distinct denominators.
Main public analysis set
422,620
85.88% of AI-related comments
Analysis-specific populations (separate eligibility rules)
Occupational lexical pool
36,713
Comments mentioning occupations
Structural comparison
4,893
Occupation-linked observations
24 eligible months
Jan 2022–Sep 2024
Precision validation
2,106
Window-specific manual annotations
Lead–lag panel
12 × 4
Occupational groups × effective periods
(b) Three-layer analytical framework
Layer 1
Descriptive analysis
Within each source
Characterize temporal and structural features.
Public discourse
Video comments
Sentiment dynamics
Topic structure
Occupation-related content
Temporal patterns
Recruitment demand
Job advertisements
Skill requirements
Occupational composition
Advertised job content
Temporal patterns
Layer 2
Cross-system comparison
Compare structural and temporal relationships across sources.
Structural correspondence
Topic–occupation alignment
Occupational distributions
Cross-source association
Temporal correspondence
Lead–lag analysis
Temporal identifiability
Alternative specifications
Composition / within-group change
Public topic strata
Recruitment occupation strata
Skill and requirement shifts
Layer 3
Integrated interpretation
Combine structural, temporal and occupational evidence without interchanging estimands.
Structural distance
Distributional difference
Precision correction
Temporal identifiability
Power from simulation
Lead–lag null distributions
Internal restructuring
Composition and within-group change
Skill / requirement shifts
Occupational heterogeneity
PSI / MRI profiles
Descriptive quadrants
Rank correlation
Robustness and sensitivity analyses
Creator-level effects
Composition and variance
Temporal granularity
Monthly / half-yearly / annual
Irony
Sensitivity to figurative language